This standard has been approved by the Naval Ship Engineering Center, Department of the Navy and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

PROCUREMENT SPECIFICATION (B) FF-S-92

P.A. SHIPS
Other Cust

SUPERSEDES:

TITLE

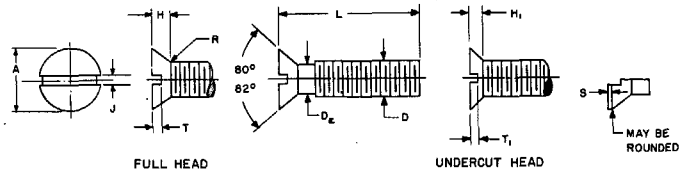
SCREWS, MACHINE, FLAT COUNTERSUNK HEAD, SLOTTED, COPPER-SILICON ALLOY, UNC-2A, NONMAGNETIC

MILITARY STANDARD

MS16218(SHIPS)

SHEET 1 OF 2

PROJECT 5305-N024Sh



FULL HEAD UNDERCUT HEAD

NOMINAL SIZE NO. OR INCH			2 (0.086)	4 (0.112)	6 (0.138)	8 (0.164)	10 (0.190)	1/4
THREAD PER INCH			56NC	40NC	32NC	32NC	24NC	20UNC
BODY DIAMETER (D_E)		MAX.	0.0860	0.1120	0.1380	0.1640	0.1900	0.2500
		MIN.	.0717	.0925	.1141	.1399	.1586	.2127
HEAD DIAMETER (A)		MAX. SHARP	.172	.225	.279	.332	.385	.507
		MIN. SHARP	.156	.207	.257	.308	.359	.477
		ABS. MIN.	.150	.200	.249	.300	.348	.462
SLOT WIDTH (J)		MAX.	.031	.039	.048	.054	.060	.075
		MIN.	.023	.031	.039	.045	.050	.064
FLAT (S)		MAX.	.003	.004	.005	.006	.007	.009
RADIUS (R)		MAX.	.036	.050	.063	.063	.083	.100
FULL HEAD	HEAD HEIGHT (H)	MAX.	.051	.067	.083	.100	.116	.153
		MIN.	.040	.055	.069	.084	.098	.131
	SLOT DEPTH (T)	MAX.	.023	.030	.038	.045	.053	.070
		MIN.	.015	.020	.024	.029	.034	.046
UNDERCUT HEAD	HEAD HEIGHT (H_1)	MAX.	.037	.048	.060	.071	.083	.108
		MIN.	.026	.036	.046	.055	.065	.086
	SLOT DEPTH (T_1)	MAX.	.019	.025	.032	.037	.044	.058
		MIN.	.015	.020	.024	.029	.034	.046

	LENGTH (L)	LENGTH TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
THREADS SHALL EXTEND TO WITHIN 2 THREADS OF THE BEARING SURFACE OF THE HEAD, OR CLOSER IF PRACTICABLE	1/ 1/8	+0 -1/32	8	18	31	46	--	--
	3/16		9	19	32	47	65	--
	1/4		10	20	33	48	66	--
	5/16		11	21	34	49	67	83
	3/8		12	22	35	50	68	84
	7/16		13	23	36	51	69	85
	1/2		14	24	37	52	70	86
	5/8		15	25	38	53	71	87
	3/4		16	26	39	54	72	88
	7/8		17	27	40	55	73	89
	1		--	28	41	56	74	90
	1-1/4	+0 -1/16	--	29	42	57	75	91
	1-1/2		--	30	43	58	76	92
	1-3/4		--	--	44	59	77	93
	2		--	--	45	60	78	94
MINIMUM COMPLETE THREAD LENGTH OF 1-3/4	2-1/4	+0 -3/32	--	--	--	61	79	95
	2-1/2		--	--	--	62	80	96
	2-3/4		---	--	--	63	81	97
	3		--	--	--	64	82	98
	4		--	--	--	--	--	179
	5		--	--	--	--	--	192

1/ HAVE UNDERCUT HEAD.

FED. SUP CLASS 5305

APPROVED 8 MAY 1957 REVISED (A) 8 ARCH 1963 REVISED (B) 8 MARCH 1967

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NOMINAL SIZE (NO. OR INCH)			5/16	3/8	7/16	1/2	5/8	3/4
THREADS PER INCH			18UNC	16UNC	14UNC	13UNC	11UNC	10UNC
BODY DIAMETER (D_E)		MAX.	0.3125	0.3750	0.4375	0.5000	0.6250	0.7500
		MIN.	.2712	.3287	.3850	.4435	.5589	.6773
HEAD DIAMETER (A)		MAX. SHARP	.635	.762	.812	.875	1.125	1.375
		MIN. SHARP	.600	.722	.771	.831	1.069	1.306
		ABS. MIN.	.581	.700	.743	.802	1.035	1.267
SLOT WIDTH (J)		MAX.	.084	.094	.094	.106	0.133	0.149
		MIN.	.072	.081	.081	.091	.116	.131
FLAT (S)		MAX.	.011	.013	.016	.018	.023	.027
RADIUS (R)		MAX.	.111	.125	.143	.154	.182	.200
FULL HEAD	HEAD HEIGHT (H)	MAX.	.191	.230	.223	.223	.298	.372
		MIN.	.165	.200	.190	.186	.253	.319
	SLOT DEPTH (T)	MAX.	.088	.106	.103	.103	.137	.171
		MIN.	.058	.070	.066	.065	.088	.111
UNDERCUT HEAD	HEAD HEIGHT (H_1)	MAX.	.135	.162	.156	.160	.212	.264
		MIN.	.109	.132	.126	.123	.167	.211
	SLOT DEPTH (T_1)	MIN.	.073	.088	.085	.085	.114	.142
		MAX.	.058	.070	.066	.065	.088	.111

	LENGTH (L)	LENGTH TOLERANCE	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.	DASH NO.
THREADS SHALL EXTEND TO WITHIN 2 THREADS OF THE BEARING SURFACE OF THE HEAD, OR CLOSER IF PRACTICABLE	1/ 1/8	+0 -1/32	--	---	---	---	---	---
	3/16		--	---	---	---	---	---
	1/4		--	---	---	---	---	---
	5/16		--	---	---	---	---	---
	3/8		99	---	---	---	---	---
	7/16		100	---	---	---	---	---
	1/2		101	114	---	---	---	---
	5/8		102	115	127	---	---	---
	3/4		103	116	128	139	---	---
	7/8		104	117	129	140	160	---
	1		105	118	130	141	161	170
	1-1/4	+0 -1/16	106	119	131	142	162	171
	1-1/2		107	120	132	143	163	172
	1-3/4		108	121	133	144	164	173
	2		109	122	134	145	165	174
MINIMUM COMPLETE THREAD LENGTH OF 1-3/4	2-1/4	+0 -3/32	110	123	135	146	166	175
	2-1/2		111	124	136	147	167	176
	2-3/4		112	125	137	148	168	177
	3		113	126	138	149	169	178
	4		180	181	193	185	188	--
	5		--	182	---	--	--	--
	5-1/2		--	183	---	--	--	--
	6		--	184	---	--	189	--
	8		--	---	---	186	190	--
	12		--	---	---	187	191	--

1/ HAVE UNDERCUT HEAD.

NOTES

1. MATERIAL: MATERIAL SHALL BE COPPER-SILICON ALLOY IN ACCORDANCE WITH QQ-C-591, COPPER ALLOY NO. 651.
2. THREADS: THREADS SHALL BE IN ACCORDANCE WITH HANDBOOK H-28.
3. *DIMENSIONS: ALL DIMENSIONS ARE IN INCHES, UNLESS OTHERWISE SPECIFIED.*
4. REFERENCED DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATION FOR BIDS.
5. FOR DESIGN FEATURE PURPOSES, THIS STANDARD TAKES PRECEDENCE OVER PROCUREMENT DOCUMENTS REFERENCED HEREIN.

APPROVED 8 MAY 1957 REVISED (B) FOR CHANGES SEE SHEETS 1 AND 2.

PROCUREMENT SPECIFICATION (B) FF-S-92

P.A. SHIPS Other Cust

SUPERSEDES:

TITLE SCREWS, MACHINE, FLAT COUNTERSUNK HEAD, SLOTTED, COPPER-SILICON ALLOY, UNC-2A, NONMAGNETIC

SHEET 2

MILITARY STANDARD

MS16218(SHIPS)

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